MEASURES OF CENTRAL TENDENCY AND LOCATION

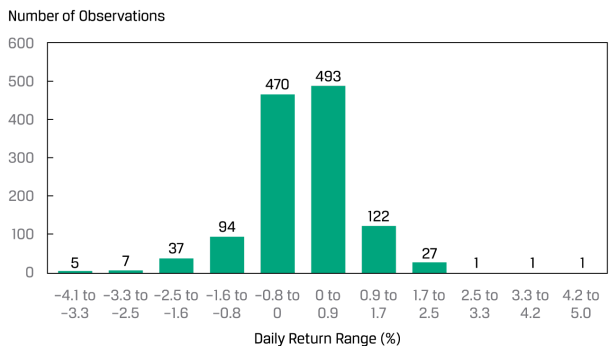
Learning Outcome

- calculate, interpret, and evaluate measures of central tendency and location to address an investment problem

In this lesson, our focus is on measures of central tendency and other measures of location. A **measure of central tendency** specifies where the data are centered. For a return series, a measure of central tendency shows where the empirical distribution of returns is centered, essentially a measure of the "expected" return based on the observed sample. **Measures of location**, mean, the **median**, and the **mode** include not only measures of central tendency but also other measures that illustrate other aspects of the location or distribution of the data.

Frequency distributions, histograms, and contingency tables provide a convenient way to summarize a series of observations on an asset's returns as a first step toward describing the data. For example, a histogram for the frequency distribution of the daily returns for the fictitious EAA Equity Index over the past five years is shown in Exhibit 1.

Exhibit 1: Histogram of Daily Returns on the EAA Equity Index



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Measures of Central Tendency

The Arithmetic Mean

Analysts and portfolio managers often want one number that describes a representative possible outcome of an investment decision. The arithmetic mean is one of the most frequently used measures of central tendency.

> ***Arithmetic Mean.*** *The **arithmetic mean** is the sum of the values of the observations in a dataset divided by the number of observations.*

The Sample Mean

The sample mean is the arithmetic mean, or arithmetic average, computed for a sample.

> ***Sample Mean Formula.*** *The **sample mean** or average, $\overline{X}$ (read "X-bar"), is the arithmetic mean value of a sample:*
>
> $$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}, \quad (1)$$
>
> *where n is the number of observations in the sample.*

A property and potential drawback of the arithmetic mean is its sensitivity to extreme values, or outliers. Because all observations are used to compute the mean and are given equal weight (i.e., importance), the arithmetic mean can be pulled sharply upward or downward by extremely large or small observations, respectively. The most common approach in this situation is to report the median, or middle value, in place of or in addition to the mean.

The Median

A second important measure of central tendency is the median.

> ***Definition of Median.*** *The median is the value of the middle item of a dataset that has been sorted into ascending or descending order. In an odd-numbered sample of n observations, the median is the value of the observation that occupies the (n + 1)/2 position. In an even-numbered sample, we define the*

Whether we use the calculation for an even- or odd-numbered sample, an equal number of observations lie above and below the median. A distribution has only one median. A potential advantage of the median is that, unlike the mean, outliers do not affect it.

The median, however, does not use all the information about the size of the observations; it focuses only on the relative position of the ranked observations. Calculating the median may also be more complex. Mathematicians express this disadvantage by saying that the median is less mathematically tractable than the mean.

The Mode

A third important measure of central tendency is the mode.

> ***Definition of Mode.*** *The mode is the most frequently occurring value in a dataset. A dataset can have more than one mode, or even no mode. When a dataset has a single value that is observed most frequently, its distribution is said to be* ***unimodal****. If a dataset has two most frequently occurring values, then it has two modes and its distribution is referred to as* ***bimodal****. When all the values in a dataset are different, the distribution has no mode because no value occurs more frequently than any other value.*

Stock return data and other data from continuous distributions may not have a modal outcome. Exhibit 2 presents the frequency distribution of the daily returns for the EAA Equity Index over the past five years.

Exhibit 2: Frequency Distribution for Daily Returns of EAA Equity Index

Return Bin (%)	Absolute Frequency	Relative Frequency (%)	Cumulative Absolute Frequency	Cumulative Relative Frequency (%)
−5.0 to −4.0	1	0.08	1	0.08
−4.0 to −3.0	7	0.56	8	0.64
−3.0 to −2.0	23	1.83	31	2.46
−2.0 to −1.0	77	6.12	108	8.59
−1.0 to 0.0	470	37.36	578	45.95
0.0 to 1.0	555	44.12	1,133	90.06
1.0 to 2.0	110	8.74	1,243	98.81
2.0 to 3.0	13	1.03	1,256	99.84
3.0 to 4.0	1	0.08	1,257	99.92
4.0 to 5.0	1	0.08	1,258	100.00

A histogram for the frequency distribution of these daily returns was shown in Exhibit 1. The modal interval always has the highest bar in the histogram; in this case, the modal interval is 0.0 to 1.0 percent, and this interval has 555 observations out of a total of 1,258 observations.

Dealing with Outliers

In practice, although an extreme value or outlier in a financial dataset may represent a rare value in the population, it may also reflect an error in recording the value of an observation or an observation generated from a different population. After having checked and eliminated errors, we can address what to do with extreme values in the sample.

When dealing with a sample that has extreme values, there may be a possibility of transforming the variable (e.g., a log transformation) or of selecting another variable that achieves the same purpose. If, however, alternative model specifications or variable transformations are not possible, then three options exist for dealing with extreme values:

Option 1: Do nothing; use the data without any adjustment.

Option 2: Delete all the outliers.

Option 3: Replace the outliers with another value.

The first option is appropriate if the values are legitimate, correct observations, and it is important to reflect the whole of the sample distribution. Because outliers may contain meaningful information, excluding or altering these values may reduce valuable information. Further, because identifying a data point as extreme leaves it up to the judgment of the analyst, leaving in all observations eliminates that need to judge a value as extreme.

The second option excludes the extreme observations. One measure of central tendency in this case is the **trimmed mean**, which is computing an arithmetic mean after excluding a stated small percentage of the lowest and highest values. For example, a 5 percent trimmed mean discards the lowest 2.5 percent and the highest 2.5 percent of values and computes the mean of the remaining 95 percent of values. A trimmed mean is used in sports competitions when judges' lowest and highest scores are discarded in computing a contestant's score.

The third option involves substituting values for the extreme values. A measure of central tendency in this case is the **winsorized mean**. It is calculated after assigning one specified low value to a stated percentage

dataset equal to the value at or below which 2.5 percent of all the values lie (as will be seen shortly, this is called the "2.5th percentile" value) and the top 2.5 percent of values in the dataset equal to the value at or below which 97.5 percent of all the values lie (the "97.5th percentile" value).

Often comparing the statistical measures of datasets with outliers included and with outliers excluded can reveal important insights about the dataset. Such comparison can be particularly helpful when investors analyze the behavior of asset returns and rate, price, spread and volume changes.

In Example 1, we show the differences among these options for handling outliers using daily returns for the fictitious Euro-Asia-Africa (EAA) Equity Index in Exhibit 2.

EXAMPLE 1

Handling Outliers: Daily Returns to an Index

Using daily returns on the EAA Equity Index for the past five years, consisting of 1,258 trading days, we can see the effect of trimming and winsorizing the data:

Exhibit 3: Effect of Trimming and Winsorizing

	Arithmetic Mean	Trimmed Mean (Trimmed 5%)	Winsorized Mean (95%)
Mean	0.035%	0.048%	0.038%
Number of Observations	1,258	1,194	1,258

The trimmed mean eliminates the lowest 2.5 percent of returns, which in this sample is any daily return less than −1.934 percent, and it eliminates the highest 2.5 percent, which in this sample is any daily return greater than 1.671 percent. The result of this trimming is that the mean is calculated using 1,194 observations instead of the original sample's 1,258 observations.

The winsorized mean substitutes a return of −1.934 percent (the 2.5 percentile value) for any observation below −1.934 and substitutes a return of 1.671 percent (the 97.5 percentile value) for any observation above 1.671. The result in this case is that the trimmed and winsorized means are higher than the arithmetic mean, suggesting the potential evidence of significant negative returns in the observed daily return distribution.

Measures of Location

Having discussed measures of central tendency, we now examine an approach to describing the location of data that involves identifying values at or below which specified proportions of the data lie. For example, establishing that 25 percent, 50 percent, and 75 percent of the annual returns on a portfolio provides concise information about the distribution of portfolio returns. Statisticians use the word **quantile** as the most general term for a value at or below which a stated fraction of the data lies. In the following section, we describe the most commonly used quantiles—quartiles, quintiles, deciles, and percentiles—and their application in investments.

Quartiles, Quintiles, Deciles, and Percentiles

We know that the median divides a distribution of data in half. We can define other dividing lines that split the distribution into smaller sizes. **Quartiles** divide the distribution into quarters, **quintiles** into fifths, **deciles** into tenths, and **percentiles** into hundredths. The **interquartile range** (IQR) is the difference between the third quartile and the first quartile, or IQR = $Q_3 - Q_1$.

Example 2 illustrates the calculation of various quantiles for the daily return on the EAA Equity Index.

EXAMPLE 2

Percentiles, Quintiles, and Quartiles for the EAA Equity Index

Using the daily returns on the EAA Equity Index over the past five years and ranking them from lowest to highest daily return, we show the return bins from 1 (the lowest 5 percent) to 20 (the highest 5 percent) as follows:

Exhibit 4: EAA Equity Index Daily Returns Grouped by Size of Return

Bin	Cumulative Percentage of Sample Trading Days (%)	Daily Return (%) between Lower Bound	Daily Return (%) between Upper Bound	Number of Observations
1	5	−4.108	−1.416	63
2	10	−1.416	−0.876	63

Bin	Cumulative Percentage of Sample Trading Days (%)	Lower Bound	Upper Bound	Number of Observations
3	15	−0.876	−0.629	63
4	20	−0.629	−0.432	63
5	25	−0.432	−0.293	63
6	30	−0.293	−0.193	63
7	35	−0.193	−0.124	62
8	40	−0.124	−0.070	63
9	45	−0.070	−0.007	63
10	50	−0.007	0.044	63
11	55	0.044	0.108	63
12	60	0.108	0.173	63
13	65	0.173	0.247	63
14	70	0.247	0.343	62
15	75	0.343	0.460	63
16	80	0.460	0.575	63
17	85	0.575	0.738	63
18	90	0.738	0.991	63
19	95	0.991	1.304	63
20	100	1.304	5.001	63

Because of the continuous nature of returns, it is not likely for a return to fall on the boundary for any bin other than the minimum (Bin = 1) and maximum (Bin = 20).

Using the data in Exhibit 4, complete the following tasks:

1. Identify the 10th and 90th percentiles.

Hide Solution

Solution:

The 10th and 90th percentiles correspond to the bins or ranked returns that include 10 percent and 90 percent of the daily returns, respectively. The 10th percentile corresponds to the return of −0.876 percent (and includes returns of that much and lower), and the 90th percentile corresponds to the return of 0.991 percent (and lower).

2. Identify the first, second, and third quintiles.

Hide Solution

Solution:

The first quintile corresponds to the lowest 20 percent of the ranked data, or −0.432 percent (and lower).

The second quintile corresponds to the lowest 40 percent of the ranked data, or −0.070 percent (and lower).

The third quintile corresponds to the lowest 60 percent of the ranked data, or 0.173 percent (and lower).

3. Identify the first and third quartiles.

Hide Solution

Solution:

The first quartile corresponds to the lowest 25 percent of the ranked data, or −0.293 percent (and lower).

The third quartile corresponds to the lowest 75 percent of the ranked data, or 0.460 percent (and lower).

4. Identify the median.

Hide Solution

Solution:

The median is the return for which 50 percent of the data lies on either side, which is 0.044 percent, the highest daily return in the 10th bin out of 20.

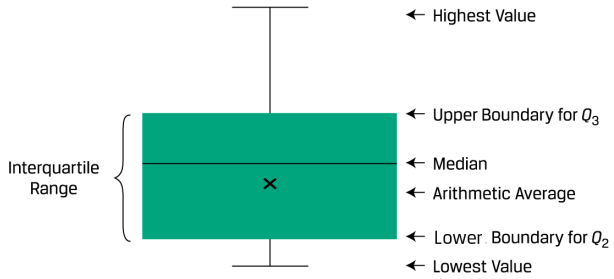
5. Calculate the interquartile range.

Hide Solution

Solution:

One way to visualize the dispersion of data across quartiles is to use a diagram, such as a box and whisker chart. A **box and whisker plot** is shown in Exhibit 5. The "box" represents the lower bound of the second quartile and the upper bound of the third quartile, with the median or arithmetic average noted as a measure of central tendency of the entire distribution. The whiskers are the lines that run from the box and are bounded by the "fences," which represent the lowest and highest values of the distribution.

Exhibit 5: Box and Whisker Plot



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There are several variations for box and whisker displays. For example, for ease in detecting potential outliers, the fences of the whiskers may be a function of the interquartile range instead of the highest and lowest values like that in Exhibit 5.

In Exhibit 5, visually, the interquartile range is the height of the box and the fences are set at extremes. But another form of box and whisker plot typically uses 1.5 times the interquartile range for the fences. Thus, the upper fence is 1.5 times the interquartile range added to the upper bound of Q_3, and the lower fence is 1.5 times the interquartile range subtracted from the lower bound of Q_2. Observations beyond the fences (i.e., outliers) may also be displayed.

We can see the role of outliers in such a box and whisker plot using the EAA Equity Index daily returns, as shown in Exhibit 6.

Exhibit 6: Box and Whisker Chart for EAA Equity Index Daily Returns

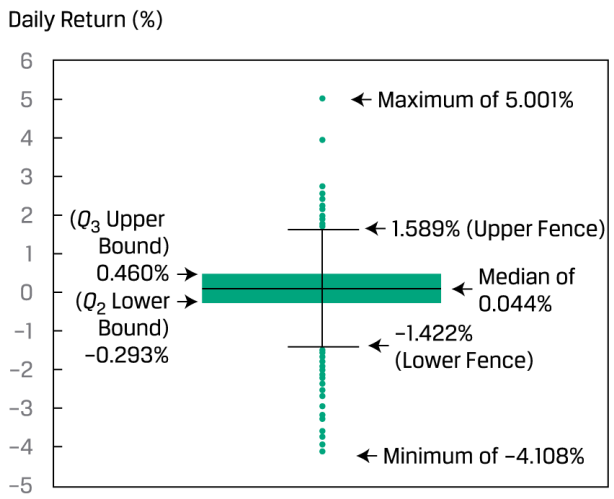


Exhibit 6 reveals the following:

- The maximum and minimum values of the distribution are 5.001 and −4.108, respectively, while the median (50th percentile) value is 0.044.
- The interquartile range is 0.753 [= 0.460 − (−0.293)], and when multiplied by 1.5 and added to the Q_3 upper bound of 0.460 gives an upper fence of 1.589 [= (1.5 × 0.753) + 0.460].
- The lower fence is determined in a similar manner, using the Q_2 lower bound, to be −1.422 [= −(1.5 × 0.753) + (−0.293)].

As noted, any observation above (below) the upper (lower) fence is deemed to be an outlier.

Quantiles in Investment Practice

Quantiles are used in portfolio performance evaluation as well as in investment strategy development and research.

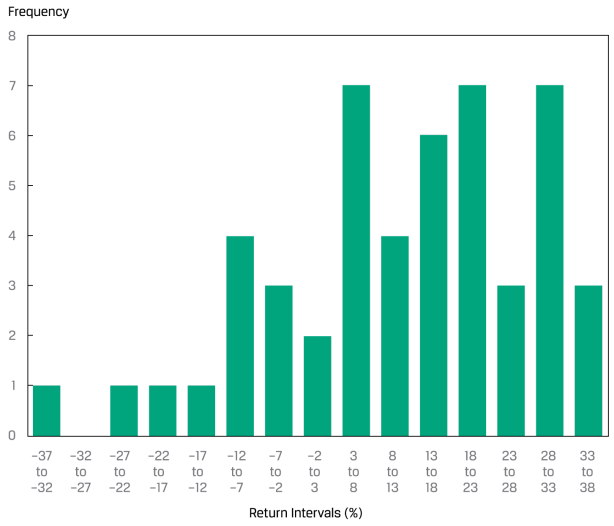
which their performance falls relative to the performance of their peer group of managers. The widely used Morningstar investment fund star rankings, for example, associate the number of stars with percentiles of performance metrics relative to similar-style investment funds.

Another key use of quantiles is in investment research. Dividing data into quantiles based on a specific objectively quantifiable characteristic, such as sales, market capitalization, or asset size allows analysts to evaluate the impact of that specific characteristic on a quantity of interest, such as asset returns, sales, growth, or valuation metrics. For instance, quantitatively driven investors often rank companies based on the market value of their equity, or their market capitalization, before sorting them into deciles. The first decile contains the portfolio of those companies with the smallest market values, usually called small capitalization companies. The tenth decile contains those companies with the largest market values, usually called large capitalization companies. Ranking companies by decile allows analysts to compare the absolute and relative performance of small market capitalization companies with large ones.

Question Set

The histogram in Exhibit 7 shows a distribution of the annual returns for the S&P 500 Index for a 50-year period.

Exhibit 7: Annual Returns for the S&P 50 Index



1. The bin containing the median return is:

A. 3 percent to 8 percent.
B. 8 percent to 13 percent.
C. 13 percent to 18 percent.

Hide Solution

Solution:

C is correct. Because 50 data points are in the histogram, the median return would be the mean of the 50/2 = 25th and (50 + 2)/2 = 26th positions. The sum of the return bin frequencies to the left of the 13 percent to 18 percent interval is 24. As a result, the 25th and 26th returns will fall in the 13 percent to 18 percent interval.

2. Based on Exhibit 7, the distribution would be *best* described as being:

A. unimodal.
B. bimodal.
C. trimodal.

Hide Solution

Solution:

C is correct. The mode of a distribution with data grouped in intervals is the interval with the highest frequency. The three intervals of 3 percent to 8 percent, 18 percent to 23 percent, and 28 percent to 33 percent all have the highest frequency of 7.

Consider the annual returns in Exhibit 8 for three portfolios for Portfolios P, Q and R. Portfolios P and R were created in Year 1, Portfolio Q was created in Year 2.

	Year 1 (%)	Year 2 (%)	Year 3 (%)	Year 4 (%)	Year 5 (%)
Portfolio P	−3.0	4.0	5.0	3.0	7.0
Portfolio Q	NA	−3.0	6.0	4.0	8.0
Portfolio R	1.0	−1.0	4.0	4.0	3.0

3. The median annual return for:

A. Portfolio P is 4.5 percent.
B. Portfolio Q is 4.0 percent.
C. Portfolio R is higher than its arithmetic mean annual return.

Hide Solution

Solution:

C is correct. The median of Portfolio R is 0.8 percent higher than the mean for Portfolio R. A is incorrect because the median annual return for Portfolio P is 4.0 percent. B is incorrect because the median annual return for Portfolio Q is 5.0 percent (midpoint of 4 percent and 6 percent).

4. The mode for Portfolio R is:

A. 1.0 percent.
B. 3.0 percent.
C. 4.0 percent.

Hide Solution

Solution:

C is correct. The mode is the most frequently occurring value in a dataset, which for Portfolio R is 4.0 percent.

A fund had the following returns over the past 10 years:

Exhibit 9: Fund Returns for 10 Years

Year	Return
1	4.5%
2	6.0%
3	1.5%
4	−2.0%
5	0.0%
6	4.5%
7	3.5%
8	2.5%
9	5.5%
10	4.0%

5. The fund's arithmetic mean return over the 10 years is *closest* to:

A. 2.97 percent.
B. 3.00 percent.
C. 3.33 percent.

Hide Solution

Solution:

B is correct. The sum of the returns is 30.0 percent, so the arithmetic mean is 30.0%/10 = 3.0%.

6. The fund's geometric mean return over the 10 years is *closest* to:

A. 2.94 percent.
B. 2.97 percent.
C. 3.00 percent.

Hide Solution

Solution:

B is correct. The geometric mean return is calculated as follows:

$$\overline{R_G} = \sqrt[10]{(1+0.045)\times(1+0.06)\times\ldots\times(1+0.055)\times(1+0.04)} - 1,$$

$$\overline{R_G} = \sqrt[10]{1.3402338} - 1 = 2.9717\%.$$

7. The harmonic mean return over the 10 years is *closest* to:

A. 2.94 percent.

Hide Solution

Solution:

A is correct. The harmonic mean is calculated as follows:

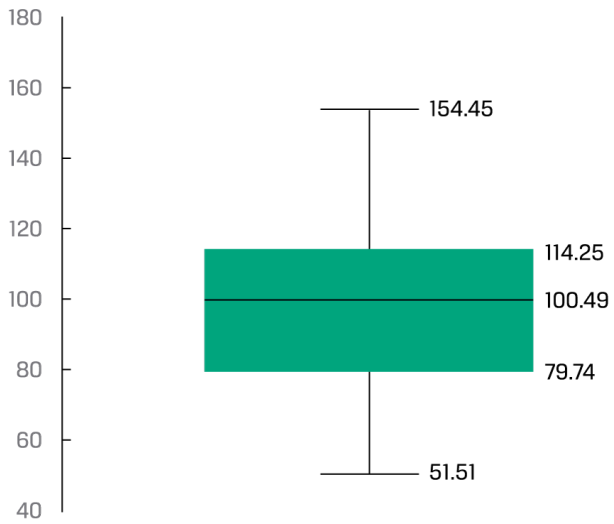
$$\overline{X_H} = \frac{n}{\sum_{i=1}^{n} (1/(1+r_i))} - 1,$$

$$\overline{X_H} = \frac{10}{\left(\frac{1}{1.045}\right)+\left(\frac{1}{1.06}\right)+\ldots+\left(\frac{1}{1.055}\right)+\left(\frac{1}{1.04}\right)} - 1,$$

$$\overline{X_H} = \left(\frac{10}{9.714}\right) - 1 = 2.9442\%.$$

Consider the box and whisker plot in Exhibit 10:

Exhibit 10: Box and Whisker Plot



8. The median is *closest* to:

A. 34.51.
B. 100.49.
C. 102.98.

Hide Solution

Solution:

B is correct. In a box and whisker plot, the "box" represents the lower bound of the second quartile and the upper bound of the third quartile, with the median or arithmetic average noted as a measure of central tendency of the entire distribution. The median is indicated within the box, which is 100.49.

9. The interquartile range is *closest* to:

A. 13.76.
B. 25.74.
C. 34.51.

Hide Solution

Solution:

C is correct. The interquartile range is the height of the box, which is the difference between 114.25 and 79.74, equal to 34.51.

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Discussion

What do you want to discuss... (required)

SC Hi all, I couldn't understand question set number 1. Could you help? thank you,

Created 4 days ago by Selarath Chum 0 replies | Last Activity: 4 days ago

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how to get the value?

Created 3 months ago by Anya Bhargava — 2 replies | Last Activity: 2 months ago
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ZN

Exhibit 6: 'The lower fence is determined in a similar manner, using the Q2 lower bound, to be −1.422 [= −(1.5 × 0.753) + (−0.293)].'
Sentence here which mentioned about 'Q2' is wrong, right? It should be Q1!

Created 4 months ago by Zongming Ni — 3 replies | Last Activity: 2 months ago
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IJ

In Exhibit 4, why is the 90th percentile value 0.991 and Not 1.304(Which is the upper range)?

Created 4 months ago by Iba Jinan Ansari — 1 reply | Last Activity: 4 months ago
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MA

The sum of the return bin frequencies to the left of the 13 percent to 18 percent interval is 24 ? how is that p

Created 4 months ago by Maimaitiming Aihemaitijiang — 0 replies | Last Activity: 4 months ago
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MO

7. The harmonic mean return over the 10 years is closest to:
A. 2.94 percent.
B. 2.97 percent.
C. 3.00 percent.

Can somebody explain to me why, following their solution I am not getting the same answer?
H.M. = 10 / [(1/1.045) + (1/1.06) + (1/1.015) + (1/0.98) + (1/1.045) + (1/1.035) + (1/1.025) + (1/1.055) + (1/1.04)] – 1
H.M. = 10 / 8.71410075373682 – 1
H.M. = 0.147565341 = 14.75653407%

Whereas the solution states the answer is B. 2.97 percent.

Going back to Quantitative Methods — Rates & Returns:
The harmonic mean is the value obtained by summing the reciprocals of the observations. The terms of the form 1/Xi, and the averaging their sum by dividing it by the number of observations, n, and, then finally, taking the reciprocal of that average.

Why in Quantitative Methods — Rates & Returns we have the HM = n / sum of 1/Xi, however in this example we have HM = n / sum of 1/(1 + ri) - 1. The other thing is we don't subtract 1 in the original teaching.

If I do it how it was taught in Quantitative Methods — Rates & Returns, I get 5.27618105%

Created 6 months ago by Maxim Ovchinnikov — 2 replies | Last Activity: 4 months ago
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SM

For Exhibit 6, how is the factor of 1.5 determined for identifying the outliers?

Created 7 months ago by Swaroop Mohapatra — 1 reply | Last Activity: 6 months ago
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